

Homework 1

- ORIE 5129. Data Science for e-Retail and the Sharing Economy.

Due: 20260911

(Question 1)

(a) The state of the system for a 2-server queueing system is the total number of customers in the system, and status of server 1 and 2.

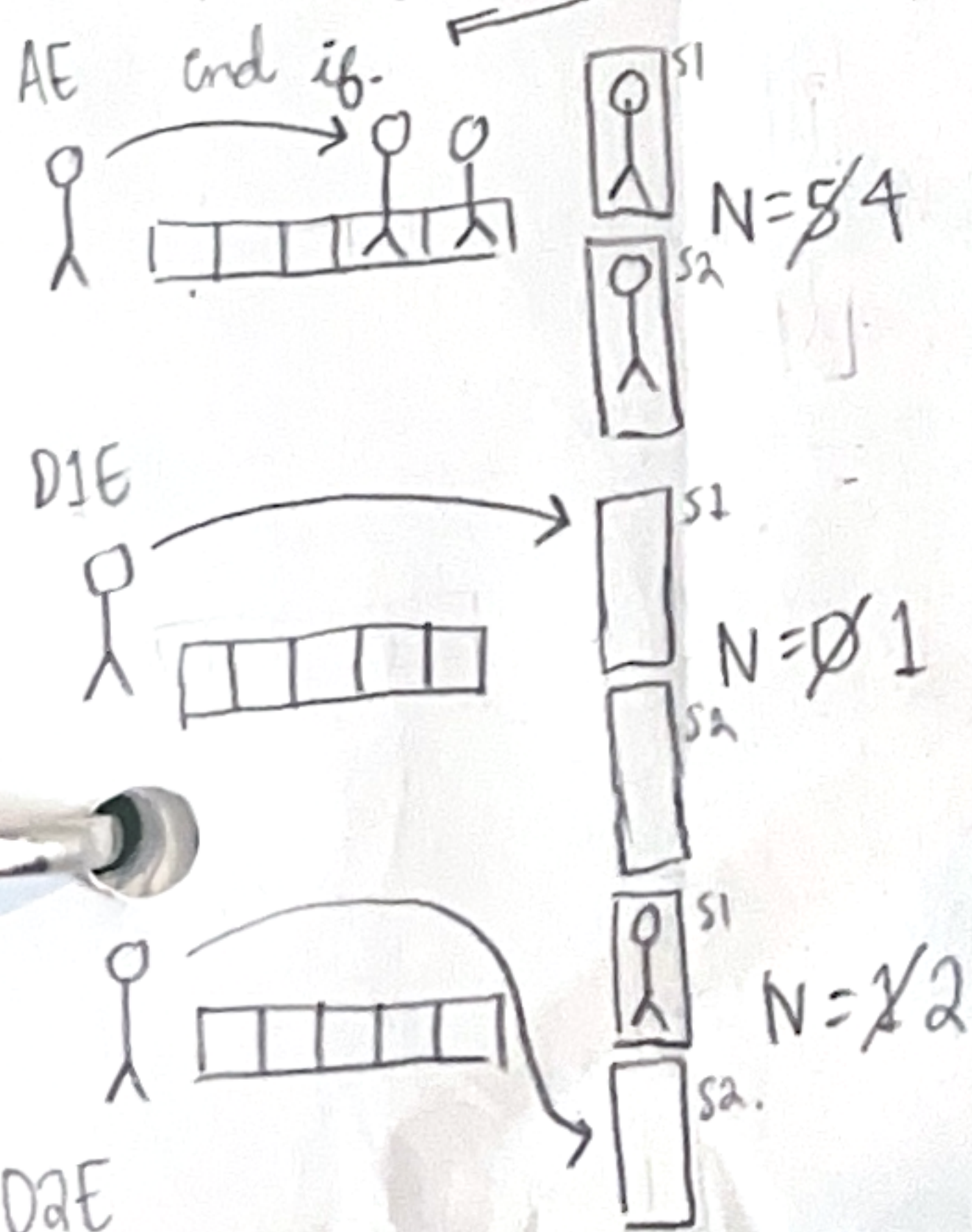
The events that change the state of the system are the

- Arrival of a new customer (AE)
- Completion of service from server 1 (D1E)
- Completion of service from server 2 (D2E)

(b) N : Total number of customers in the system

Say we are at time t and arrival event AE occurs.

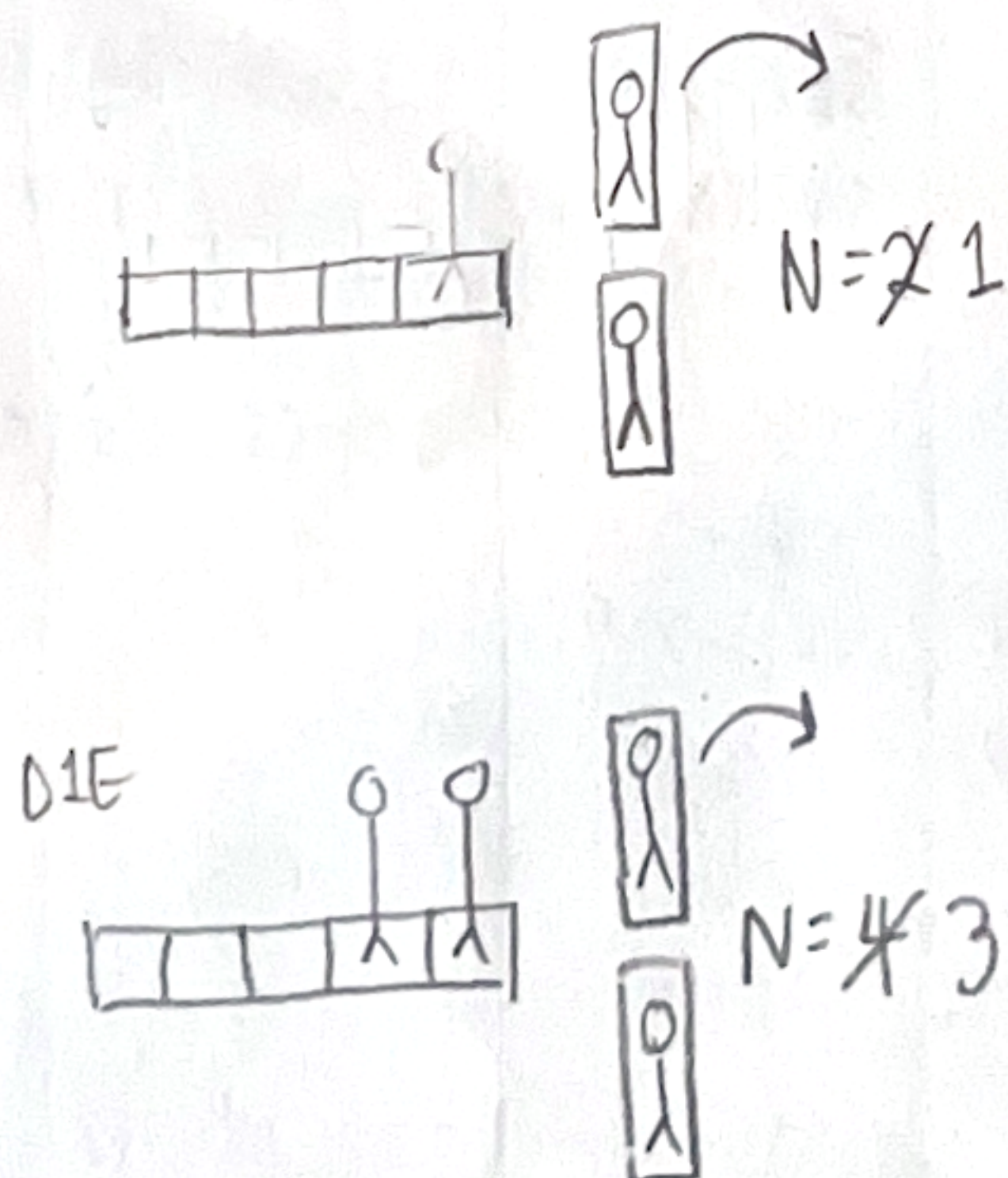
- $N++$
- Sample interarrival time IA
- Schedule arrival event AE at time $t+IA$
- **if** ($N==1$) **or** ($S1==0$) **then**
 - Sample service time for server 1, $ST1$
 - Schedule departure event D1E at time $t+ST1$
 - $S1=1$
- else** ($S2==0$) **then**
 - Sample service time for server 2, $ST2$
 - Schedule departure event D2E at time $t+ST2$
 - $S2=1$



Say we are at time t and departure event D1E occurs

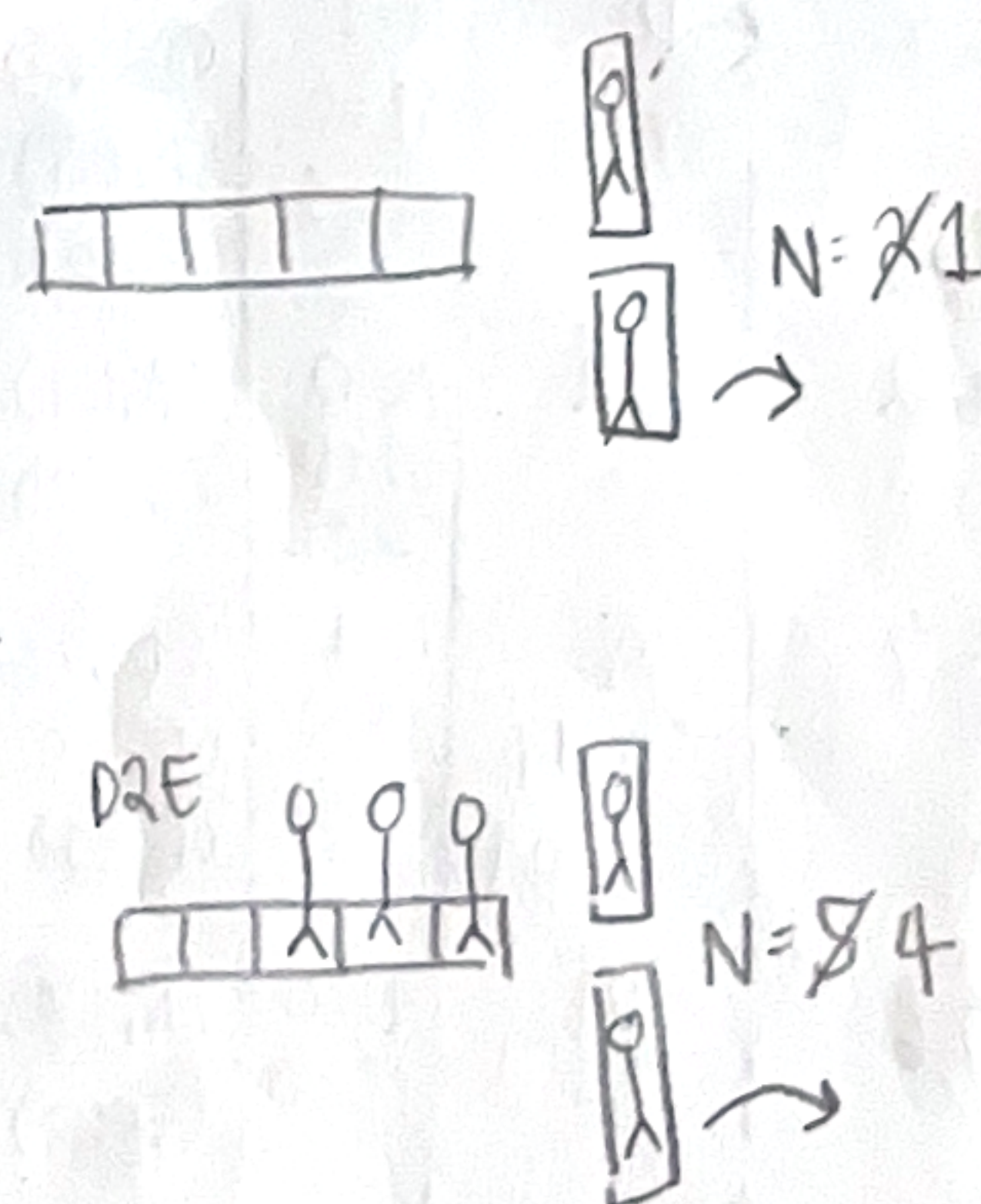
- $N--$
- $S1=0$
- **if** ($N>1$) **then**
 - Sample service time for server 1, $ST1$
 - Schedule departure event D1E at time $t+ST1$
- end if**

* Can also compare N to $S2$



Say we are at time t and departure event D2E occurs

- $N--$
- $S2=0$
- **if** ($N>1$) **then**
 - Sample service time for server 2, $ST2$
 - Schedule departure event D2E at time $t+ST2$
 - $S2=1$
- end if**



(c) $IA = \{1.3, 1.1, 0.6, 1.5, 2.2, 1.1, 0.8, 0.7, 0.1, 1.4, 1.2, 1.4, 1.4, 1.2, 2.1\}$

$ST1 = \{2.1, 1.3, 0.9, 1.4, 1.2, 2.1, 2.2\}$

$ST2 = \{2.2, 1.6, 1.1, 1.7, 2.1, 1.3, 0.9\}$

Time	Event	State of Sys	FEL	Total Time S1 Busy	Total Time Both Busy	Stickera	Total time by arrivals	# of arrivals
0	Begin sim $IA = 1.3$	$N=0$ $S1=0$ $S2=0$	(A, 1.3) (E, 10)	0	0	$S1(-)$ $S2(-)$ $Q(-)$	0	0
1.3	A $IA = 1.1$ $ST1 = 2.1$	$N=1$ $S1=1$ $S2=0$	(A, 2.4) (D1, 3.4) (E, 10)	0	0	$S1(1.3)$ $S2(-)$ $Q(-)$	0	1
2.4	A $IA = 0.6$ $ST2 = 2.2$	$N=2$ $S1=1$ $S2=1$	(A, 3.0) (D1, 3.4) (D2, 4.6) (E, 10)	1.1	0	$S1(1.3)$ $S2(2.4)$ $Q(-)$	0	2
3.0	A $IA = 1.5$	$N=3$ $S1=1$ $S2=1$	(D1, 3.4) (A, 4.5) (D2, 4.6) (E, 10)	1.7	0.6	$S1(1.3)$ $S2(2.4)$ $Q(3.6)$	0	3
3.4	D1 $ST1 = 1.3$	$N=2$ $S1=1$ $S2=1$	(A, 4.5) (D2, 4.6) (D1, 4.7) (E, 10)	2.1	1.0	$S1(3.0)$ $S2(2.4)$ $Q(-)$	$0 + (3.4 - 1.3)$ $= 2.1$	3
4.5	A $IA = 2.2$	$N=3$ $S1=1$ $S2=1$	(D2, 4.6) (D1, 4.7) (A, 6.7) (E, 10)	3.2	2.1	$S1(3.0)$ $S2(2.4)$ $Q(4.5)$	2.1	4
4.6	D2 $ST2 = 1.6$	$N=2$ $S1=1$ $S2=1$	(D1, 4.7) (D2, 6.2) (A, 6.7) (E, 10)	3.3	2.2	$S1(3.0)$ $S2(4.5)$ $Q(-)$	$2.1 + (4.6 - 2.4)$ $= 4.3$	4
4.7	D1	$N=1$ $S1=0$ $S2=1$	(D2, 6.2) (A, 6.7) (E, 10)	3.4	2.3	$S1(-)$ $S2(4.5)$ $Q(-)$	$4.3 + (4.7 - 3.0)$ $= 6.0$	4
6.2	D2	$N=0$ $S1=0$ $S2=0$	(A, 6.7) (E, 10)	3.4	2.3	$S1(-)$ $S2(-)$ $Q(-)$	$6.0 + (6.2 - 4.5)$ $= 7.7$	4
6.7	A $IA = 1.1$ $ST1 = 0.9$	$N=1$ $S1=1$ $S2=0$	(D1, 7.6) (A, 7.8) (E, 10)	3.4	2.3	$S1(6.7)$ $S2(-)$ $Q(-)$	7.7	5
7.6	D1	$N=0$ $S1=0$ $S2=0$	(A, 7.8) (E, 10)	4.3	2.3	$S1(-)$ $S2(-)$ $Q(-)$	$7.7 + (7.6 - 6.7)$ $= 8.6$	5

Time	Event	State of Sys	FEL	Total Time S1 Busy	Total Time Both Busy	Stickers	Total time by arrival	# of arrivals
7.8	A IA=0.5 ST1=1.4	N=1 S1=1 S2=0	(A, 8.3) (D1, 9.2) (E, 10)	4.3	2.3	S1(7.8) S2(-) Q(-)	8.6	6
8.3	A IA=0.7 ST2=1.1	N=2 S1=1 S2=1	(A, 9.0) (D1, 9.2) (D2, 9.4) (E, 10)	4.8	2.8	S1(7.8) S2(8.3) Q(-)	8.6	7
9.0	A IA=0.1	N=3 S1=1 S2=1	(A, 9.1) (D1, 9.2) (D2, 9.4) (E, 10)	5.5	3.5	S1(7.8) S2(8.3) Q(9.0)	8.6	8
9.1	A IA=1.4	N=4 S1=1 S2=1	(D1, 9.2) (D2, 9.4) (E, 10) (A, 10.5)	5.6	3.6	S1(7.8) S2(8.3) Q(9.1, 9.0)	8.6	9

- Proportion of time first server is busy = $\frac{\text{Total time first server is busy}}{\text{Total time}}$

$$= \frac{6.5}{10.0} = \boxed{0.65}$$

- Proportion of time both servers are busy = $\frac{\text{Total time both are busy}}{\text{Total time}}$

$$= \frac{4.5}{10.0} = \boxed{0.45}$$

- Average amount of time arriving customer spends in system.

$$= \frac{\int_0^T N(t) dt}{\text{Total \# arriving customers}} = \frac{13}{9} = \boxed{1.44 \text{ minutes}}$$

9.2	D1 ST1=1.2	N=3 S1=1 S2=1	(D2, 9.4) (E, 10) (A, 10.5) (D1, 10.4)	5.7	3.7	S1(9.0) S2(8.3) Q(9.1)	8.6 + (9.2 - 7.8) = 10	9
9.4	D2 ST2=1.7	N=2 S1=1 S2=1	(E, 10) (D1, 10.4) (A, 10.5) (D2, 11.1)	5.9	3.9	S1(9.0) S2(9.1) Q(-)	10 + (9.4 - 8.3) = 11.1	9
10	E	N=2 S1=1 S2=1	(...)	6.5	4.5	S1(9.0) S2(9.1) Q(-)	11.1 + (10 - 9.0) + (10 - 9.1) = 13	9